

Cross-Asset Price Discovery at Two Frequencies: Findings from the Corrected Replication

Memo to co-authors — R. Garrison, August 11, 2026

Sample: 10 volatile + 10 paired baseline sessions (2022–2026) + the four March-2020 MWCB sessions; SPY consolidated NBBO/ladder vs front-month ES from the CME venue ladder; one vendor pull at 10ms with the 1s frames derived exactly from it. All estimates apply the halt-mask policy (halt snapshots and reopen seams excluded from every estimator), the fixed-(1, -1) VECM with per-day `ec_valid` screening, and day-clustered or permutation inference throughout. Stack version v0.9.67.

One housekeeping note before the findings: the four stage-report files are two identical pairs — the two 1s reports match byte-for-byte (analysis lags auto-scale to 5 at 1s) and the two 10ms reports match (auto-scale to 60). The 15-vs-60 lag contrast lives in the Table 9 system, whose `table9_both_ways_*.csv` outputs you have now sent; Section 7 analyzes that experiment. Sections 1–6 are the **two-frequency comparison** (1s vs 10ms), which turns out to be an equally interesting axis.

1. Summary of findings

1. **Futures leadership is conditional, not unconditional.** On the full 24-session panel the working paper’s headline — ES dominates price discovery — does not survive as an unconditional claim: at 1s the component share of ES is 0.38 and its information share 0.48 (near parity, SPY slightly ahead); at 10ms ES regains a modest lead (IS 0.54, CS 0.46). But **in the stressed regime ES is clearly ahead at both frequencies**, and the volatile-minus-benchmark gap is significant at the 5% level by day permutation at both grids. The futures market leads exactly when the paper’s mechanism — tandem trading under extraordinary volatility — is operating. This is a sharper and, we think, more defensible claim than unconditional dominance.
2. **Tandem trading survives the corrected nulls at 1s and is directly measurable at the innovation level — but the action-time claim does not survive.** Against independence *given the observed marginals*, the corner log-odds ratio rises monotonically from baseline (1.10) to volatile (1.39) to MWCB (1.47). The measured correlation of SPY and ES order-flow *innovations* is 0.74 on benchmark days and 0.83 on volatile days. However, at the frequency-matched binomial null, the observed off-corner mass is only 1.32× the null at 10ms and 0.97× in action time (vs 33.5× at 1s); the published Table 7 comparison of all three rows against the per-second null overstates the fine-frequency evidence by orders of magnitude.
3. **The liquidity mechanism is a fine-grid phenomenon.** The state-dependent error-correction loading — the paper’s “arbitrage weakens when the book thins” channel — is sharply identified at 10ms ($t = 5.2$ on the mid, $t = 11.7$ on the microprice) and statistically invisible at 1s ($t = 0.25/1.35$). Equilibrium half-life on stressed books is roughly three minutes.

4. **Discontinuities are where the futures lead is largest.** At 10ms, ES’s information share in the jump component (0.56) exceeds its continuous-component share (0.39), and on 2020-03-09 the co-jump tape shows ES leading SPY into common jumps 7,610 times against 1,669 (4.6:1) with 94.8% sign agreement.
5. **Two of the paper’s identification devices need restating.** The Hasbrouck bounds at 1s are nearly vacuous ([0.00, 1.00]-style) and tighten dramatically at 10ms — the fine grid is the right place to quote IS. And Rigobon heteroskedasticity identification fails its own pre-test on this sample (variance-ratio spread $0.055 < 0.15$): the regimes differ in scale, not in relative heteroskedasticity, so the Cholesky bracket — not het-ID — is what we can honestly report.

2. Price discovery across frequencies and regimes

Full-panel means (24 sessions, all `ec_valid`):

Exhibit 1. Full-panel price-discovery shares by sampling frequency.

	1s	10ms
mean IS_{ES} (midpoint of Hasbrouck bounds)	0.484	0.542
mean CS_{ES} (Gonzalo–Granger)	0.384	0.455
days with invalid error correction	0 of 24	0 of 24

The 2014–2017 sample’s futures dominance does not reappear unconditionally on 2022–2026 data. Two things changed relative to the working paper’s estimates: the halt windows no longer contribute pseudo-returns (the four MWCB days previously entered with the 900-second halt and its reopen seam included), and the ES leg is now the venue ladder with the price-level replay validated against it. At 1s, SPY carries slightly more than half of the common trend on the average day.

The regime split — the paper’s central comparison — survives at both grids:

Exhibit 2. ES component share by regime, with day-level permutation p -values.

CS_{ES}	benchmark	volatile	difference	permutation p (day-level)
1s	0.269	0.466	+0.197	0.048
10ms	0.387	0.504	+0.117	0.047

ES’s share of the permanent component rises materially in the stressed regime at both frequencies. The pooled fixed-effects panel VECM (lags built within-day, day fixed effects, day-clustered SEs) shows the same sign — at 10ms the volatile-interaction t -statistics are -1.50 (SPY) and -1.60 (ES), $p \approx 0.13$ – 0.15 — but with only 24 day-clusters the pooled interaction does not clear conventional thresholds. With $N = 24$ days, the day-level permutation test is the appropriately sized test, and it rejects at both grids. We recommend reporting both: the permutation p as the headline, the clustered panel t as the honest pooled counterpart.

Why to quote the 10ms information shares. At 1s the Hasbrouck bounds are close to uninformative on most days — e.g., 2023-08-07 gives $IS_{ES} \in [0.004, 0.861]$ — because at that sampling interval nearly all SPY–ES adjustment is contemporaneous and the ordering assumption does all the work. At 10ms the same day’s bounds are $[0.244, 0.523]$. Asynchrony at the fine grid breaks the simultaneity; the bounds become

estimates rather than restatements of the ordering. This materially strengthens the paper’s measurement section.

3. Tandem order flow against the corrected nulls

The published Table 5 rejected a Binomial($n, \frac{1}{2}$) null that bundles “each market is a fair coin” with “the markets are independent”; the marginals alone reject it. Against independence *conditional on the observed marginals* — the null that isolates cross-market trading — the dependence is still decisively present and **increases with stress**:

Exhibit 3. Tandem order flow against the marginal-preserving independence null (revised Table 5).

Panel	$P_{\text{CMOF}}/\text{indep.}$	log-odds ratio	corner asymmetry
A. Baseline	1.27	1.098	−0.014
B. Volatile	1.34	1.385	−0.008
C. MWCB	1.37	1.470	−0.019
C'. MWCB ex-SSR	1.36	1.413	−0.021

The corner asymmetry (sell-side minus buy-side local log odds) is approximately zero in every panel: no Rule 201 fingerprint at the pooled level, so the dependence is symmetric tandem trading rather than mechanically constrained selling. (2020-03-16 was short-sale-restricted all session and is reported both ways.)

The frequency-matched null changes Table 7’s message. The per-second null (0.4% per corner) is not portable across aggregations, because the null depends entirely on per-bar order counts:

Exhibit 4. Off-corner mass against the frequency-matched null (revised Table 7).

Aggregation	observed off-corners	null at actual counts	ratio
1 second	11.9%	0.35%	33.5
10ms	30.1%	22.9%	1.32
action time	48.4%	50.1%	0.97

The 48.4% action-time figure that reads as overwhelming against 0.4% is almost exactly its own null. The defensible statement is: tandem dependence is strong and highly significant at the one-second aggregation, attenuates toward the null as bars shrink to the arrival scale, and the *level* comparison across frequencies in the published table should be replaced by per-frequency ratios.

Tandem flow at the innovation level. The correlation of SPY and ES OFI innovations is 0.742 (benchmark) and 0.835 (volatile) — direct, model-free evidence of common flow, rising under stress. This has a knock-on consequence for the variance decomposition: the orthogonal FEVD, which assumes uncorrelated flow shocks, attributes 95% of ES return variance to “ES flow”; the generalized (Pesaran–Shin) decomposition at the measured correlation attributes 65% to ES flow and 35–43% to SPY flow, with the caveat that under tandem

flow part of each share is common flow counted toward both shocks. We suggest the GFEVD as the quotable decomposition, presented explicitly as an upper bound on separability.

4. Liquidity-conditional price discovery and the ECM-SDE

The paper’s mechanism — price discovery migrates and error correction weakens when liquidity withdraws — shows up strongly at the fine grid and weakly at 1s:

Exhibit 5. The state-dependent error-correction loading (a_1 , SPY leg) across frequencies.

	1s	10ms
t -statistic (mid)	0.25	5.24
t -statistic (microprice)	1.35	11.70
median IS_{ES} along the liquidity curve (mid/microprice)	0.21/0.34	0.67/0.74
implied equilibrium half-life (stressed states)	$\sim 10s$	$\sim 3min$

At 10ms, moving along the relative-liquidity state shifts the adjustment burden exactly as the paper argues: κ falls and the half-life lengthens as the book thins, and the ES share of discovery along the entire curve sits at 0.58–0.63 (0.62–0.63 on the microprice). The depth-state and FPCA-state day-by-day splits are heterogeneous in sign day to day — the pooled interacted estimator, not the per-day quantile splits, is the exhibit to lead with.

Cross-impact confirms the direction of causation in stress. Within each grid, the average cross-impact coefficient roughly triples from benchmark to volatile days (10ms: 0.020 $\rightarrow$ 0.060; 1s: 0.198 $\rightarrow$ 0.268), and the asymmetry runs the right way for the paper’s thesis: on the MWCB days the impact of ES flow on SPY returns (0.13–0.25) is two to five times the impact of SPY flow on ES returns (0.02–0.06). Futures flow moves the ETF; the reverse channel is an order of magnitude weaker.

5. Jumps and co-jumps

At 10ms with the Lee–Mykland local-volatility classifier, 58% of common-factor quadratic variation on the average day is discontinuous (the truncation estimate at 1s is 17%, and 6% under LM — the fine grid is where jumps are measurable at all). Two results:

- **ES leads more at the discontinuities than in the diffusion:** mean $IS_{j,ES} = 0.563$ against $IS_{c,ES} = 0.390$. Price discovery’s jump component is a futures phenomenon.
- **Co-jump timing** (2020-03-09): of 24,346 common jumps, ES moves first 7,610 times, SPY first 1,669 times (4.6:1), 15,067 simultaneous at 10ms, with 94.8% sign agreement.

This is, in our view, the single sharpest exhibit for futures leadership the revision can offer — it is model-free, it is at the events that matter, and it does not depend on a Cholesky ordering.

6. Identification and inference upgrades the revision should adopt

1. **Day-clustered and permutation inference.** The legacy pooled-iid t on the SPY-return $\sim$ ES-flow regression is 253.3; day-clustered it is 4.62 (wild-cluster bootstrap $p = 0.001$). The result survives, the stars change; a referee will insist, and we now do it everywhere (24 clusters; Webb weights at the MWCB $G = 4$).
2. **Rigobon het-ID fails its pre-test on this sample.** The variance-ratio spread across regimes is 0.055 (threshold 0.15): the regimes scale both legs' variances nearly equally, so the rotation is unidentified and the het-ID point estimates are numerical noise. Report the two Cholesky orderings as a bracket and the het-ID verdict row; do not quote the het-ID coefficients. (With MWCB days as their own third regime the over-identification statistic is available as a specification test — worth one row in the appendix.)
3. **The realized SPY–ES correlation is 0.932; the DCC persistence is $a + b = 0.9999$.** The conditional-correlation path is near-integrated on this sample; level statements about “correlation rising in stress” are safer made with per-bar realized correlation (non-overlapping bars) than with the DCC path.
4. **Lag robustness at the analysis layer.** The VECM-based shares barely move across lag choices at 1s (CS_{ES} 0.384–0.412 for $p \in \{3, 5, 10, 20\}$) — the lag sensitivity that plagues the correlation system (the Eq. (5) SVAR, where BIC chases the correlation window's own MA structure to the search bound, our mechanical restatement of footnote 17) does not afflict the price-discovery estimates. The Table 9 lag question is exactly where your 15-vs-60 comparison belongs; the RealBar and DCC dependent variables are the window-free columns to quote there. Section 7 runs that comparison on your two outputs.
5. **Book-state beats the quoted spread as the liquidity conditioner:** R^2 for |SPY returns| 0.318 vs 0.138, with partial R^2 0.179 for the state given the spread — the paper's liquidity narrative strengthens under the richer state variable.

7. Table 9 at 15 versus 60 lags: the window artifact made visible

Your two STAGE 5 runs estimate the same Eq. (5) correlation system on the same data — 1s grid, 100-bar rolling window, fixed-effects panel VAR (lags built within-day, day fixed effects, day-cluster bootstrap SEs, Romano–Wolf joint stars) — differing only in the imposed lag depth, VAR(15) versus VAR(60); the 60-lag run also carries the DCC column. This is the cleanest demonstration we have of the point in §6.4.

The rolling-window columns are not lag-robust. The Pearson impact responses shrink by a factor of three to more than ten between $p = 15$ and $p = 60$ (several cells collapse to zero at the reported precision), and the Romano–Wolf star pattern reshuffles: ten Pearson cells carry stars in at least one run, only two keep them in both ($RV_{ES}/\text{volatile}$, $0.342 \rightarrow 0.031$, and $WtdSpread_{SPY}/\text{benchmark}$, $0.092 \rightarrow 0.027$), seven of the nine starred at $p = 15$ lose them at $p = 60$, and $OFI_{ES}/\text{benchmark}$ is starred only at $p = 60$. HY, which shares the rolling window, shrinks the same way ($RV_{ES}/\text{volatile}$ $0.134 \rightarrow 0.029$; $WtdSpread_{ES}/\text{benchmark}$ $-0.138 \rightarrow -0.031$). The Epps-artifact share is itself lag-dependent: the HY–Pearson delta is nearly three-quarters of the published-design response at $p = 15$ ($RV_{ES}/\text{benchmark}$: -0.380 against 0.517) and about a quarter at $p = 60$ (-0.030 against 0.130).

Exhibit 6. The published (Pearson) design at two lag depths: one- σ orthogonalized impact responses of Δ -correlation $\times 100$. Day-cluster bootstrap SEs in parentheses; Romano–Wolf joint stars (***/**/* = 1/5/10% FWER). MicroDev rows omitted (never significant).

shock	volatile, $p=15$	volatile, $p=60$	benchmark, $p=15$	benchmark, $p=60$
Spread _{ES}	−0.014** (0.004)	−0.000 (0.001)	−0.012 (0.006)	0.002 (0.001)
WtdSpread _{ES}	0.073* (0.026)	0.000 (0.001)	−0.060*** (0.011)	−0.001 (0.005)
OFl _{ES}	0.005 (0.006)	0.000 (0.002)	−0.021 (0.009)	−0.014*** (0.003)
RV _{ES}	0.342** (0.094)	0.031** (0.008)	0.517*** (0.053)	0.130 (0.053)
Spread _{SPY}	−0.028 (0.011)	0.002 (0.001)	−0.043*** (0.012)	0.002 (0.003)
WtdSpread _{SPY}	0.061*** (0.015)	0.002 (0.002)	0.092*** (0.016)	0.027** (0.007)
OFl _{SPY}	0.003 (0.003)	0.001 (0.001)	−0.007** (0.002)	0.002 (0.001)
RV _{SPY}	0.144 (0.060)	−0.018 (0.012)	0.176 (0.064)	0.056 (0.036)

The mechanism is the one we identified analytically: with a 100-bar rolling window the dependent variable is a moving-average object of order $\approx W$, so at $p = 15$ nearly all of that structure sits in the residual — the one- σ orthogonalized impact responses are measured against a residual that still contains the window — while at $p = 60$ much of it has been absorbed and the yardstick changes. Since $p < W$ in both runs, neither magnitude is the “right” one, and a criterion left to choose p simply chases W (the BIC-at-the-bound result, our mechanical restatement of footnote 17).

The window-free column is sign-stable but not magnitude-stable. RealBar responses scale up roughly three- to seven-fold going to $p = 60$, with standard errors moving in the same direction, so magnitudes are not comparable across lag depths in this column either. But the inference is far more stable: the three cells starred at both depths are the same three (WtdSpread_{ES}/volatile, OFl_{ES}/benchmark, RV_{ES}/benchmark), no starred cell changes sign between runs, and no cell is significant with opposite signs in the two runs.

Exhibit 7. The window-free (RealBar) column at two lag depths: Δ Fisher-z of the 60s-bar non-overlapping realized correlation $\times 100$. MicroDev rows omitted (never significant).

shock	volatile, $p=15$	volatile, $p=60$	benchmark, $p=15$	benchmark, $p=60$
Spread _{ES}	0.372 (0.159)	1.521 (0.630)	−0.313*** (0.057)	−0.578 (0.502)
WtdSpread _{ES}	0.960*** (0.160)	5.412*** (0.729)	−0.077 (0.107)	0.456 (0.617)
OFl _{ES}	−0.083 (0.090)	0.711 (0.419)	−0.317** (0.078)	−1.538*** (0.370)
RV _{ES}	0.281 (0.167)	6.106*** (1.013)	1.604*** (0.322)	11.750*** (0.808)
Spread _{SPY}	−0.010 (0.251)	0.984 (0.683)	0.157 (0.237)	1.051 (0.754)
WtdSpread _{SPY}	0.329 (0.139)	0.810 (0.776)	0.393* (0.138)	1.255 (0.452)
OFl _{SPY}	0.033 (0.103)	1.249*** (0.309)	−0.010 (0.091)	−0.387 (0.423)
RV _{SPY}	0.717** (0.205)	1.973 (0.738)	0.210 (0.561)	1.480 (0.569)

What survives everywhere — the quotable core. Across both lag depths and all measurement designs: (i) volatility shocks raise subsequent correlation — RV_{ES} is the only shock starred in Pearson, HY, and RealBar in both runs; (ii) book-liquidity (weighted-spread) shocks are, with RV, the only other shocks that stay jointly significant across designs and depths, with a regime-dependent sign pattern — the ES-side response is positive in the volatile regime and negative on benchmark days in every starred cell, while the SPY-side response is positive where starred; (iii) ES order-flow-imbalance shocks lower benchmark-day correlation in the window-free column at both depths ($-0.317^{**} \rightarrow -1.538^{***}$); (iv) microprice-deviation shocks do nothing anywhere. The DCC column (60-lag run only) agrees in miniature: its starred cells are the weighted-spread rows and $RV_{SPY/benchmark}$, with the smallness and window-independence expected of a recursive filter.

Recommendation. Publish Table 9 from the window-free columns (RealBar, with DCC as corroboration), present it as a sign-and-significance exhibit rather than a magnitudes exhibit, fix the lag depth ex ante on the bar grid (where the selection criterion is not chasing the window), and keep the Pearson-vs-HY pair in the appendix as the demonstration that the published design’s magnitudes were window artifacts to first order — which is precisely what this 15-vs-60 experiment shows. Orthogonalized impact responses are denominated in the size of each equation’s innovation, and deepening the lag polynomial re-sizes those innovations in every column; signs and joint significance are the transportable content of this system.

8. Sample and data caveats for the appendix

- **Rolls.** 2020-03-18: calendar pick ESM0 carries 69.5% of two-contract volume; 2024-12-18: ESH5 carries 64.2%. Report the measured shares; do not splice (10–12 point calendar-spread seams). **2025-06-13: the calendar rule picked the minority contract** (ESU5, 13.3% of volume) — we must either re-extract pinned to ESM5 and say so, or keep the rule and report the share; it cannot pass silently.
- **2020-03-16** is Rule-201 restricted all session (reported in and out of the MWCB panel).
- **Halt masking** excludes 900 halt seconds per MWCB session plus the reopen seams from every estimator; the four 2020 sessions’ estimates are on the $\sim 22.5k$ (1s)/2.25M (10ms) tradable observations.
- **Staleness at 10ms** (79.6% zero-return snapshots for SPY, 89.2% for ES): second moments at the fine grid come from noise-robust estimators and the fleeting-quote filter is engaged on flow inputs; per-bar realized correlation, not tick-by-tick Pearson, is the fine-grid correlation object.
- One open validation item: the independent message-replay vs venue-ladder cross-check on ESH5 2024-12-18 recorded a disagreement we have not yet diagnosed; until then the ladder validation exhibit should stay out of the draft.

9. What we would change in the paper, concretely

1. Recast “futures dominate price discovery” as “**futures leadership is a stress phenomenon**”: near parity (1s) to modest ES lead (10ms) unconditionally; +12 to +20 points of common-trend share in the volatile regime (permutation $p < 0.05$ at both grids); largest at the discontinuities (jump IS 0.56; co-jump lead 4.6:1).
2. Keep Table 5 with the marginal-preserving null and the log-odds ratio (monotone in stress); **replace Table 7’s cross-frequency level comparison with per-frequency ratios** and let the action-time row say

what it now says.

3. Quote information shares at 10ms (tight bounds), with 1s as the robustness column.
4. Lead the mechanism section with the interacted ECM-SDE at 10ms ($t = 5.2/11.7$, three-minute stressed half-lives) and the cross-impact asymmetry.
5. Adopt day-cluster/wild-cluster/permutation inference throughout; retire the pooled-iid stars; report the Rigobon verdict row and the Cholesky bracket.
6. Add the innovation-level tandem correlation ($0.74 \rightarrow 0.83$ in stress) as the direct measurement of the paper's title phenomenon, and the GFEVD as its variance-accounting consequence.
7. Rebuild Table 9 on the window-free dependent variables (RealBar, DCC) as a sign-and-significance exhibit with the lag depth fixed ex ante, and move the Pearson-vs-HY contrast to the appendix as the measurement-artifact demonstration (§7).

Next: the ESH5 validation diagnosis (validate_ESH5_20241218.txt), and the 2025-06-13 minority-contract decision.